



Internship Outline

3 Months

Program Overview

The **CYBERSTER Internship** is a **12-week, globally structured cybersecurity internship** designed for **beginners to early-intermediate learners** seeking **industry-aligned, entry-level readiness** in offensive, defensive, and hybrid security roles.

This internship is built around a **practical, responsibility-driven learning framework**, where interns learn by executing real tasks rather than attending routine classes. The program follows a structured progression model:

Concept → Demonstration → Guided Practice → Independent Execution

This approach ensures participants develop not only technical understanding, but also **professional discipline, autonomy, and real-world problem-solving skills** expected in modern cybersecurity roles.

Internship Structure

- **Duration:** 12 Weeks (3 Months)
- **Format:**
 - **1 Masterclass per Week**
 - Used exclusively to **assign weekly tasks**,
 - **explain expectations**,
 - and **guide interns** on execution, reporting, and improvement.

No daily theory or lab classes are conducted. Interns are expected to **work independently**, following weekly objectives and timelines—replicating real internship and industry environments.

Participants are assigned to one of three specialized teams:

- **RED Team** – Offensive Security & Penetration Testing
- **BLUE Team** – Defensive Security, Monitoring & Incident Response



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- **PURPLE Team** – Integrated Red–Blue Operations

Each team follows a **clear, role-specific roadmap** aligned with real-world cybersecurity workflows and professional practices.

Work Model & Evaluation

Throughout the internship, participants complete:

- **Weekly Assigned Tasks** aligned with their team role
- **Guided Exercises** explained during the weekly masterclass
- **Independent Practical Execution** to build self-reliance
- **Progress Tracking & Performance Evaluation** based on consistency, quality, and professionalism

Interns progress through a **simulated security lifecycle** relevant to their assigned team, developing both technical execution skills and structured professional thinking.

Recognition & Opportunities

Upon successful completion, interns receive:

- **Certificate of Completion**
- **Internship Offer Letter**

In addition, the **Top 10 performers from each team (Red, Blue, and Purple)** are awarded a **Recommendation Letter**, based on performance quality, consistency, and demonstrated capability throughout the program.

Program Vision

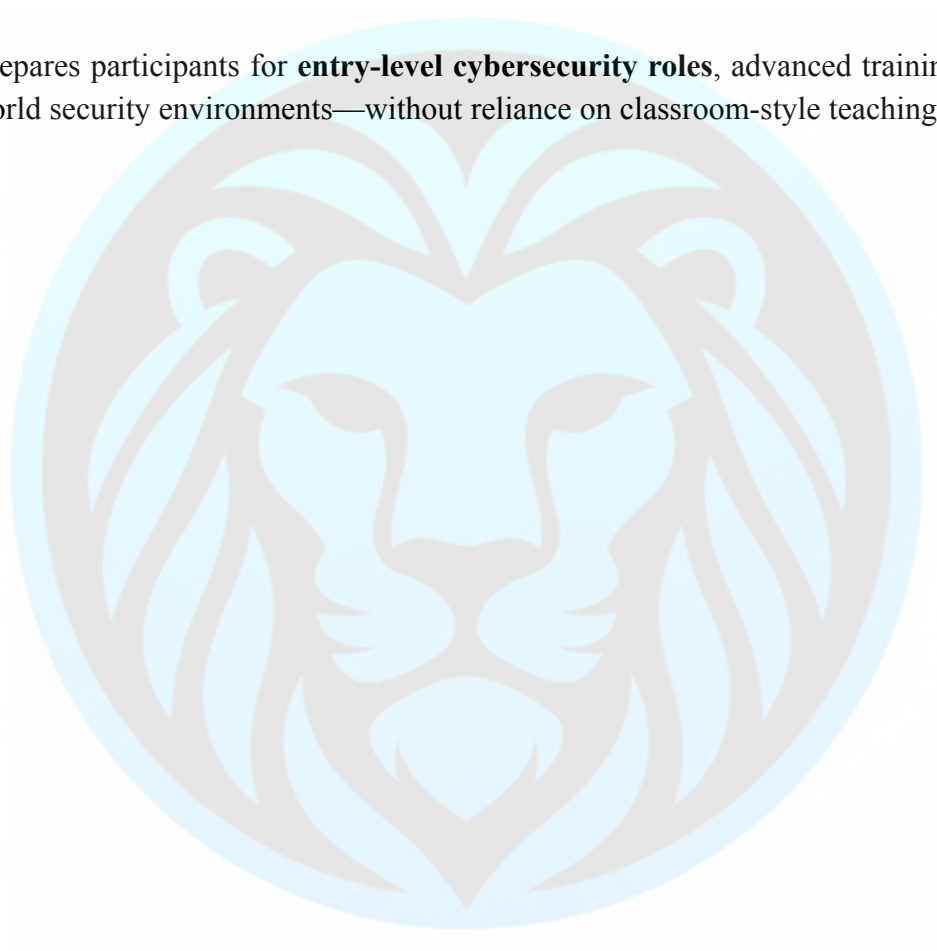
The CYBERSTER Internship is designed to reflect **international internship standards**, emphasizing **hands-on responsibility, guided mentorship, and professional execution**. The



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program prepares participants for **entry-level cybersecurity roles**, advanced training pathways, and real-world security environments—without reliance on classroom-style teaching.





Red Team Outline

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Month 1 for Red Team

Week 01: Advanced Reconnaissance & Attack Surface Mapping

Objective: To master the art of information gathering and identifying the digital footprint of a target without being detected.

Task 01: Passive OSINT & Subdomain Enumeration

- **Description:** Conduct a thorough passive reconnaissance on a given target domain (e.g., a bug bounty target or a dummy corporate domain).
- **Technical Requirements:**
 - Perform subdomain discovery using a combination of tools: **Subfinder**, **Assetfinder**, and **Amass**.
 - Use **DNS Dumpster** and **CRT.sh** to find historical subdomains and SSL certificate leaks.
 - Identify "hidden" assets that are not indexed by search engines.
- **Learning Goal:** Understand that no single tool is enough; cross-referencing results is key to a successful Red Team operation.

Task 02: Infrastructure & Technology Profiling

- **Description:** Identify the technologies powering the target's infrastructure.
- **Technical Requirements:**
 - Use **WhatWeb** or the **Wappalyzer CLI** to identify CMS, Web Servers, and Frameworks.
 - Perform **advanced Google Dorking** to find sensitive files like `phpinfo()`, `.env`, `robots.txt`, and backup directories (`.bak`, `.zip`).
 - Use **WHOIS** and **NSLookup** to map the IP ranges and DNS records belonging to the organization.



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- **Learning Goal:** Mapping the tech stack helps in narrowing down potential exploits in later weeks.

Task 03: GitHub Dorking & Credential Leaks

- **Description:** Search for leaked secrets and hardcoded credentials in public repositories.
- **Technical Requirements:**
 - Use **GitHacker** or manual GitHub search queries to look for keywords like "API_KEY", "PASSWORD", "AWS_SECRET", or "SQL_CONNECTION" related to the target organization.
 - Analyze commit histories to find credentials that were deleted but are still present in previous versions.
- **Learning Goal:** Experience how human error and developer negligence provide the easiest entry points for attackers.

Task 04: Visual Reconnaissance & Live Asset Identification

- **Description:** Filter out dead assets and visualize the live attack surface.
- **Technical Requirements:**
 - Use **Httpprobe** or **Httpx** to verify which subdomains identified in Task 01 are actually alive.
 - Capture screenshots of all live web assets using **Aquatone** or **Gowitness** to identify interesting login panels or dev environments.
- **Learning Goal:** Save time by focusing only on active, high-value targets.

Weekly Deliverable: The Reconnaissance Report

1. **Executive Summary:** High-level overview of the target's attack surface.
2. **Asset List:** A clean CSV or table of all discovered subdomains and IPs.
3. **Technology Map:** A list of frameworks and services found.



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4. **Security Gaps:** Any initial "Low-Hanging Fruit" found during OSINT (e.g., exposed `.git` folders).

Week 02: Network Enumeration & Service Vulnerability Discovery

Objective: To transition from broad reconnaissance to targeted network scanning, identifying open ports, and fingerprinting services while avoiding basic security triggers.

Task 01: Advanced Network Scanning with Nmap

- **Description:** Move beyond basic ping sweeps to conduct deep-dive port scanning.
- **Technical Requirements:**
 - Perform **TCP Connect scans** (`-sT`) and **SYN Stealth scans** (`-sS`) and compare the results in terms of speed and visibility.
 - Conduct **UDP Scanning** (`-sU`) on the top 100 most common UDP ports (DNS, SNMP, DHCP).
 - Use **Timing Templates** (`-T0` to `-T5`) to understand how scan speed affects reliability and detection.
- **Learning Goal:** Understand the mechanics of the TCP 3-way handshake and why Stealth scans are preferred in professional Red Teaming.

Task 02: Service Fingerprinting & Banner Grabbing

- **Description:** Identify the exact versions of services running on open ports to look for known vulnerabilities.
- **Technical Requirements:**
 - Use Nmap's **Version Detection** (`-sV`) and **Aggressive Scan** (`-A`) flags to identify service details.



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- Perform **Manual Banner Grabbing** using **Netcat (nc)** or **Telnet** to verify service headers.
- Use **Enum4linux** or **Samba tools** if SMB (Port 445) is found open to list shares and users.
- **Learning Goal:** Realize that a service version is the primary "key" to finding a specific exploit (CVE).

Task 03: Leveraging Nmap Scripting Engine (NSE)

- **Description:** Use automated scripts to detect vulnerabilities and misconfigurations during the scan.
- **Technical Requirements:**
 - Run scripts from the '**Discovery**' and '**Safe**' categories to find additional info (e.g., http-enum, ssl-cert).
 - Execute the '**Vuln**' category scripts (--script vuln) on a target machine to identify exploitable vulnerabilities.
 - Map the NSE output to specific **CVE IDs**.
- **Learning Goal:** Automate the repetitive parts of vulnerability discovery while maintaining control over the scan.

Task 04: Stealth & Firewall Evasion Techniques

- **Description:** Practice techniques to bypass basic Intrusion Detection Systems (IDS) or Firewalls.
- **Technical Requirements:**
 - Use **Packet Fragmentation** (-f) and MTU manipulation to slip past simple packet filters.
 - Experiment with **Decoys** (-D) to hide your actual IP address among a crowd of "fake" scanning IPs.
 - Use **Source Port Spoofing** (--source-port) to bypass firewalls that only allow traffic from specific ports (like DNS/53).



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- **Learning Goal:** Learn that "Loud" scans get blocked; "Quiet" and "Creative" scans get results.

Weekly Deliverable: The Scanning & Enumeration Report

1. **Nmap Output Analysis:** Screenshots and explanations of the open ports found.
2. **Service Table:** A clean table listing: Port, Service Name, Version, and Potential Vulnerabilities.
3. **Evasion Results:** A short write-up on which stealth flags worked and which were potentially "noisy."
4. **Actionable Insights:** Which port/service is the most likely "Initial Access" point and why?

Week 03: Vulnerability Assessment & Initial Exploitation

Objective: To master the process of mapping discovered services to known vulnerabilities (CVEs) and performing the first stage of a successful exploit to gain a foothold.

Task 01: Advanced Web Directory & Parameter Discovery

- **Description:** Find the hidden "Entry Points" of a web application.
- **Technical Requirements:**
 - Use **ffuf**, **Dirsearch**, or **Gobuster** to perform directory brute-forcing using the **SecLists** wordlist.
 - Identify hidden directories (e.g., `/admin`, `/config`, `/dev`, `.git`).
 - Use **Arjun** or **ParamSpider** to find hidden URL parameters (e.g., `?id=`, `?file=`, `?page=`).
- **Learning Goal:** Understand that web vulnerabilities are often hidden in non-indexed pages and parameters.



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Task 02: CVE Mapping & Exploit Research

- **Description:** Take the service versions discovered in Week 02 and find working exploits for them.
- **Technical Requirements:**
 - Use the **Searchsploit CLI** tool to find local exploit code for specific service versions.
 - Research vulnerabilities on the **NVD** and **Exploit-DB**.
 - Focus on **Remote Code Execution (RCE)** and **Foothold** exploits.
- **Learning Goal:** Learn how to verify if a vulnerability is actually exploitable in a real-world scenario.

Task 03: CMS Vulnerability Scanning (WordPress/Joomla)

- **Description:** Identify weaknesses specific to CMS platforms.
- **Technical Requirements:**
 - Use **WPScan** on a WordPress target to identify vulnerable plugins and **Username Enumeration**.
 - Use **JoomScan** if the target is running Joomla.
 - Look for outdated plugins that are common entry points.
- **Learning Goal:** Understand that third-party plugins are often the weakest link.

Task 04: The First Shell - Metasploit Basics

- **Description:** Use the Metasploit Framework to launch your first successful exploit.
- **Technical Requirements:**
 - Use **msfconsole** to search for modules related to the CVEs found in Task 02.
 - Set up **PAYLOAD**, **LHOST**, and **RHOST** correctly.
 - Execute an exploit against a vulnerable VM (Metasploitable 2 or 3).
 - **Bonus:** Use a **Netcat listener** (`nc -lvnp`) to catch a "Reverse Shell" manually.



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- **Learning Goal:** Experience the transition from a "scanner" to a "hacker" by gaining an interactive session.

Weekly Deliverable: The Vulnerability & Foothold Report

1. **Vulnerability Matrix:** Found CVEs, severity (CVSS), and exploit source.
2. **Web Discovery Log:** Screenshots of hidden directories.
3. **Exploitation Proof:** Screenshot of whoami or hostname output from the shell.
4. **Remediation:** One-sentence prevention advice for every exploit used.

Week 04: Web Application Penetration Testing (OWASP Top 10)

Objective: To identify and exploit the most common web-based vulnerabilities using industry-standard tools and manual techniques.

Task 01: Burp Suite Mastery (The Interceptor)

- **Description:** Focuses on traffic manipulation.
- **Technical Requirements:**
 - Configure **Burp Suite** with your browser (using **FoxyProxy**).
 - Practice **Intercepting** and modifying requests in the Proxy tab.
 - Use **Repeater** to replay requests and **Intruder** for simple dictionary attacks.
- **Learning Goal:** Understand the "Request-Response" cycle and how to manipulate data.

Task 02: Injection Attacks (SQLi & Command Injection)

- **Description:** Trick the database and the operating system through input fields.
- **Technical Requirements:**



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- Perform a **Manual SQL Injection** to extract database version/user.
- Use **SQLMap** to automate table and column extraction.
- Test for **OS Command Injection** using commands like `;` `whoami` or `&& dir`.
- **Learning Goal:** Learn why "Input Validation" is the most critical defense.

Task 03: Broken Access Control & IDOR

- **Description:** Test if you can access data that doesn't belong to you by changing identifiers.
- **Technical Requirements:**
 - Perform an **IDOR** test: Change `user_id` in a request to access another user's data.
 - Test for **Privilege Escalation**: Can a "Viewer" become an "Admin" by changing a cookie value (e.g., `role=user` to `role=admin`)?
- **Learning Goal:** Understand that logic flaws are often more dangerous than technical bugs.

Task 04: Cross-Site Scripting (XSS) & File Inclusion

- **Description:** Execute code in the user's browser and read internal server files.
- **Technical Requirements:**
 - Find a **Reflected XSS** vulnerability and trigger `alert(document.cookie)`.
 - Test for **Local File Inclusion (LFI)** by reading `/etc/passwd` or `win.ini`.
 - Understand the risk of **Remote File Inclusion (RFI)**.
- **Learning Goal:** See how client-side attacks lead to session hijacking and server-side attacks to data disclosure.

Weekly Deliverable: The OWASP Top 10 Audit Report

1. **Vulnerability Breakdown:** "Steps to Reproduce" guide for each bug.
2. **Impact Analysis:** Explanation of what an attacker could achieve.
3. **Proof of Concept (PoC):** High-quality screenshots of successful exploits.
4. **Remediation:** Provide the "Code-level Fix" (e.g., Prepared Statements).



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Month 02: Internal Network & Privilege Escalation

Week 05: Post-Exploitation & Privilege Escalation

- **Objective:** To learn how to elevate privileges from a low-level user to a Root/System administrator and establish a permanent presence on the target machine.
- **Task 01: Linux Privilege Escalation (The Path to Root)**
 - **Description:** You have a low-privilege shell (e.g., www-data). Now, you must become root.
 - **Technical Requirements:**
 1. **Manual Enumeration:** Check for misconfigured SUDO rights (sudo -l), SUID bits (find / -perm -u=s -type f 2>/dev/null), and world-writable files in /etc/.
 2. **Automated Enumeration:** Use scripts like LinPEAS.sh and LSE.sh to identify potential escalation paths.
 3. **Kernel Exploits:** Research the OS version and identify if it's vulnerable to exploits like DirtyPipe or PwnKit (Only as a last resort).
 - **Learning Goal:** Understand that most "hacks" happen due to poor system administration rather than complex bugs.
- **Task 02: Windows Privilege Escalation (The Path to SYSTEM)**
 - **Description:** Elevate privileges on a Windows environment from a standard user to NT AUTHORITY\SYSTEM.
 - **Technical Requirements:**
 1. **Enumeration:** Use WinPEAS.exe and PowerUp.ps1 to find unquoted service paths, weak service permissions, and stored credentials in the registry.



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- 2. **Token Impersonation:** Learn how to use tools like PrintSpoofer or the "Potato" family (Juicy/God/Rotten Potato) if the user has `SeImpersonatePrivilege`.
 - **Learning Goal:** Mastering Windows-specific misconfigurations like "Unquoted Service Paths."
- **Task 03: Establishing Persistence (Don't Lose the Shell)**
 - **Description:** Ensure that you can get back into the system even if the victim reboots or the connection drops.
 - **Technical Requirements:**
 - 1. **Linux:** Create a Cron job that sends a reverse shell every hour, or add your own SSH public key to the `~/.ssh/authorized_keys` file.
 - 2. **Windows:** Create a Scheduled Task or add a malicious entry to the Registry Run Keys (`HKCU\Software\Microsoft\Windows\CurrentVersion\Run`).
 - **Learning Goal:** Understand how real-world APTs (Advanced Persistent Threats) stay inside a network for months.
- **Task 04: Post-Exploit Enumeration & Data Exfiltration**
 - **Description:** Find the "Crown Jewels"—passwords, config files, and sensitive data.
 - **Technical Requirements:**
 - 1. Search for configuration files (e.g., `web.config`, `settings.py`, `.env`) that might contain database passwords.
 - 2. **Credential Harvesting:** Dump browser-stored passwords or search for SSH private keys (`id_rsa`).
 - 3. **Exfiltration:** Transfer a small sensitive file from the target to your machine using SCP, Python3 HTTP server, or Netcat.
 - **Learning Goal:** Remember that "Data" is the ultimate goal of any attack, not just the shell.
- **Weekly Deliverable: The Post-Exploitation Report**
 - **Requirement:** Submit a professional PDF report containing:
 - 1. **Escalation Path:** Detailed steps (with screenshots) of how you moved from a low-privilege user to Admin/Root.



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2. **Tools Used:** A list of automated scripts and manual commands that provided the most value.
3. **Persistence Proof:** A screenshot showing that your persistence method survived a system reboot.
4. **Security Recommendation:** How can the administrator patch the specific misconfiguration you used? (e.g., "Fixing SUID permissions on binary X").

Week 06: Lateral Movement & Network Pivoting

- **Objective:** To master the art of moving through an internal network after the initial breach and reaching "isolated" machines that are not directly connected to the internet.
- **Task 01: Internal Network Discovery (Living off the Land)**
 - **Description:** Use the compromised machine to find other targets in the same internal network.
 - **Technical Requirements:**
 1. **Network Mapping:** Use built-in OS tools like `arp -a`, `netstat -ano`, and `ipneighbor` to identify other active IPs in the subnet.
 2. **Scanning from the Inside:** Upload a static version of `nmap` or use a simple Bash/PowerShell loop to scan for open ports (like 80, 445, 3389) on neighboring internal IPs.
 - **Learning Goal:** Understand that you shouldn't always upload heavy tools; using what's already on the system is quieter and more professional.
- **Task 02: Pivoting & Tunneling (The "Double Jump")**
 - **Description:** Set up a tunnel so your attack machine (Kali) can talk to a second internal machine through the first compromised host.
 - **Technical Requirements:**
 1. **SOCKS Proxy:** Set up a SOCKS proxy using SSH Dynamic Port Forwarding (`ssh -D`) or tools like Chisel.
 2. **Proxychains:** Configure `proxychains` on your Kali machine to route your tools (like `Nmap` or `Burp Suite`) through the established tunnel.



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- 3. **Metasploit Autoroute:** Practice using the autoroute post-module and SOCKS4a server inside Metasploit.
 - **Learning Goal:** Mastering the concept of "Routing" traffic through a compromised gateway to reach deep internal targets.
- **Task 03: Credential Dumping & Harvesting**
 - **Description:** Extract passwords and hashes from the first machine to use them for the second machine.
 - **Technical Requirements:**
 1. **Windows (Mimikatz):** Use Mimikatz (or its PowerShell version) to dump LSA secrets, SAM hashes, and clear-text passwords from memory.
 2. **Linux:** Extract hashes from /etc/shadow and search for "Configuration Files" (like wp-config.php or .env) that contain hardcoded database credentials.
 3. **Hash Cracking:** Use John the Ripper or Hashcat to crack the harvested hashes locally on your Kali machine.
 - **Learning Goal:** Learn that "Passwords found on Machine A often work on Machine B."
- **Task 04: Lateral Movement Techniques**
 - **Description:** Use the harvested credentials to gain access to a second internal machine.
 - **Technical Requirements:**
 1. **Pass-the-Hash (PtH):** Use Impacket tools (like psexec.py or wmiexec.py) to log into a second Windows machine using only a hash—no password required.
 2. **Remote Desktop (RDP):** Use xfreerdp with captured credentials to gain GUI access to an internal target.
 3. **WinRM/SSH:** Use Evil-WinRM (Windows) or SSH (Linux) to move laterally across the network.
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- **Learning Goal:** Understand that "Initial Access" is only the beginning; lateral movement is how you reach the "Domain Controller" or the "Database."
- **Weekly Deliverable: The Lateral Movement Report**
 - **Requirement:** Submit a technical PDF report containing:
 1. **Network Diagram:** A simple drawing showing your Kali machine, the "Pivot Host" (Machine 1), and the "Internal Target" (Machine 2).
 2. **Pivoting Steps:** Commands used to set up the tunnel and the proxy.
 3. **Captured Evidence:** Screenshots of the dumped credentials and the successful login to the second machine.
 4. **Security Recommendation:** How to prevent lateral movement (e.g., "Network Segmentation" or "Disabling LLMNR/NBT-NS").

Week 07: Active Directory Enumeration & Kerberos Attacks

- **Objective:** To understand the architecture of a Windows Domain environment and learn how to identify misconfigurations in the Kerberos authentication protocol.
- **Task 01: Understanding AD Architecture**
 - **Description:** Before attacking, you must understand the components of a Windows Domain.
 - **Technical Requirements:**
 1. **Theory:** Research and document the roles of a Domain Controller (DC), Global Catalog, Organizational Units (OU), and Group Policy Objects (GPO).
 2. **Protocol Research:** Explain the difference between LDAP, SMB, and RPC in an AD context.



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- **Learning Goal:** You cannot hack what you don't understand. Mapping the "Forest" and "Trees" is the first step.
- **Task 02: AD Enumeration (The "Quiet" Way)**
 - **Description:** Query the Domain Controller to find users, groups, and computers without triggering alerts.
 - **Technical Requirements:**
 1. **PowerView:** Use Get-NetUser, Get-NetComputer, and Get-NetGroup to list domain assets.
 2. **BloodHound:** Run SharpHound.exe to collect AD data and import it into the BloodHound GUI (Kali) to visualize attack paths (e.g., finding which user has Admin rights on which machine).
 3. **LDAP Search:** Use ldapsearch or ADEplorer to browse the directory tree manually.
 - **Learning Goal:** Identifying the "High-Value Targets" (Domain Admins) and the paths to reach them.
- **Task 03: AS-REP Roasting (Attacking Users without Pre-Auth)**
 - **Description:** Find users who do not require Kerberos pre-authentication and steal their password hashes.
 - **Technical Requirements:**
 1. Identify users with the property DONT_REQ_PREAUTH enabled.
 2. Use Impacket-GetNPUsers to request an AS-REP for these users.
 3. **Cracking:** Take the output hash and attempt to crack it using Hashcat (Mode 18200).
 - **Learning Goal:** Exploiting a common misconfiguration in user account settings to gain initial domain credentials.
- **Task 04: Kerberoasting (Attacking Service Accounts)**
 - **Description:** Request service tickets for Service Principal Names (SPNs) and crack them offline to get service account passwords.
 - **Technical Requirements:**
 1. Use GetUserSPNs.py (Impacket) or Rubeus to list accounts with SPNs.
 2. Request a TGS (Ticket Granting Service) ticket and export the hash.



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- 3. **Cracking:** Use John the Ripper or Hashcat (Mode 13100) to crack the service account hash.
- **Learning Goal:** Service accounts often have weak passwords and high privileges; Kerberoasting is the #1 way to escalate privileges in AD.
- **Weekly Deliverable: The AD Enumeration & Kerberos Report**
 - **Requirement:** Submit a technical PDF report containing:
 1. **AD Environment Map:** A list of discovered Domain Controllers and high-privilege groups.
 2. **BloodHound Graph:** A screenshot of a "Shortest Path to Domain Admin" graph.
 3. **Attack Proof:** Screenshots of successful AS-REP Roasting and Kerberoasting (showing the hashes).
 4. **Security Recommendation:** How to defend against these attacks (e.g., "Enforcing Kerberos Pre-authentication" and "Using Long/Complex Passwords for Service Accounts").

Week 08: AD Domain Dominance & Advanced Persistence

- **Objective:** To move from a standard domain user to a Domain Administrator and establish high-level persistence that allows the Red Team to maintain control over the entire Windows Forest.
- **Task 01: LLMNR/NBT-NS Poisoning (The Initial Credential Theft)**
 - **Description:** Intercept network traffic to steal password hashes from users who are trying to access non-existent network shares.
 - **Technical Requirements:**
 1. Use Responder (responder -I eth0 -dwv) to listen for LLMNR, NBT-NS, and MDNS requests.
 2. Capture NTLMv1/v2 hashes from the network.
 3. **Cracking:** Use Hashcat (Mode 5600) to crack the captured NTLMv2 hashes.
 - **Learning Goal:** Understand how legacy protocols in Windows environments are a goldmine for attackers.



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- **Task 02: Exploiting Weak ACLs & GPO Abuse**
 - **Description:** Abuse permissions granted to users to escalate to Domain Admin or modify Group Policies.
 - **Technical Requirements:**
 1. **ACL Abuse:** Use BloodHound to find users with GenericWrite or AllExtendedRights over high-value objects (like a Domain Admin user or a Group).
 2. Use Add-DomainObjectAcl (PowerView) to grant yourself "DCSync" rights.
 3. **GPO Abuse:** Use SharpGPOAbuse to create a malicious GPO that adds your user to the local Administrators group on all workstations.
 - **Learning Goal:** Learn that misconfigured permissions are often more dangerous than software bugs in Active Directory.
- **Task 03: The Golden & Silver Ticket Attacks**
 - **Description:** Forge Kerberos tickets to gain permanent, undetected access to services and the entire Domain.
 - **Technical Requirements:**
 1. **Golden Ticket:** Extract the KRBTGT account hash using Mimikatz (DCSync). Use this hash to forge a TGT that gives you Domain Admin rights for 10 years.
 2. **Silver Ticket:** Forge a TGS ticket for a specific service (like CIFS or HTTP) to access a specific server without touching the Domain Controller.
 3. **Tooling:** Practice using Mimikatz (command: kerberos::golden) or Rubeus.
 - **Learning Goal:** Understand the "God Mode" of AD Hacking and why protecting the KRBTGT account is the #1 priority for Blue Teams.
- **Task 04: DCSync & NTDS.dit Extraction**
 - **Description:** Mimic a Domain Controller to pull all user hashes from the database.
 - **Technical Requirements:**



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1. Use Impacket's secretsdump.py or Mimikatz to perform a DCSync attack.
2. Dump the entire NTDS.dit database to get the NTLM hashes of every single user in the organization.
3. Perform Pass-the-Hash (PtH) using a dumped Administrator hash to verify access.
 - **Learning Goal:** Learn the final step of a total network takeover.
- **Weekly Deliverable: The Domain Dominance Report**
 - **Requirement:** Submit a professional PDF report containing:
 1. **Attack Chain:** A step-by-step flow from how you got the first hash (Responder) to how you became Domain Admin (DCSync).
 2. **Artifact Analysis:** Show the forged Golden Ticket and explain how it works.
 3. **Persistence Strategy:** How would you maintain access if the Admin changes their password?
 4. **Security Hardening:** Provide 3 critical steps to defend against AD attacks (e.g., "Implementing Tiered Administration," "Disabling LLMNR," and "Rotating KRBGT passwords twice").

Month 3 for Red Team

Month 03 | Week 09: AV/EDR Evasion & Defensive Bypassing

Objective: To learn how to bypass modern security defenses like Windows Defender, AMSI, and basic EDR solutions using obfuscation and "Living off the Land" techniques.

Task 01: Payload Obfuscation & Signature Evasion



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- **Description:** Standard payloads (like MSFVenom) are easily caught by Antivirus. You must learn to hide them.
- **Technical Requirements:**
 - Generate a standard meterpreter shell and check it against **VirusTotal** (or a private scanner to avoid distribution).
 - Use **Encoders** (like shikata_ga_nai) and compare detection rates.
 - **Advanced:** Use **Packers** (like UPX) or **Crypters** to wrap the payload.
 - Use tools like **Msfvenom-Payload-Creator (MPC)** or **Villain** to create stealthier connections.
- **Learning Goal:** Understand how signature-based detection works and how simple changes in the file structure can bypass it.

Task 02: Bypassing AMSI (Antimalware Scan Interface)

- **Description:** AMSI is a Windows feature that scans scripts (PowerShell, VBScript) in memory. You must learn to disable it.
- **Technical Requirements:**
 - Research and execute an **AMSI Bypass** script in PowerShell (e.g., using AmsiScanBuffer memory patching).
 - Test if your PowerShell commands are still being flagged after the bypass.
 - Use **Obfuscated PowerShell** tools like **Invoke-Obfuscation** to hide your intent from logging.

Learning Goal: Learn why "In-Memory" attacks are more effective than writing files to the disk.

Task 03: LOLBAS (Living Off The Land Binaries and Scripts)

- **Description:** Instead of uploading malware, use trusted Windows binaries to perform malicious actions.
- **Technical Requirements:**
 - Use **Certutil.exe** to download a file from a remote server (bypassing browser/firewall alerts).



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- Use **Regsvr32.exe** (Squiblydoo attack) to execute a remote script without it ever touching the disk.
- Explore the **LOLBAS Project** website and identify 3 more binaries that can be used for execution or persistence.
- **Learning Goal:** Discover how to hide "In Plain Sight" by using the operating system's own tools against itself.

Task 04: Process Injection & Hollowing

- **Description:** Hide your malicious code inside a legitimate running process (like explorer.exe or notepad.exe).
- **Technical Requirements:**
 - Use a tool or a custom script (C++ or C#) to perform **Process Injection**.
 - Practice **Process Hollowing**: Start a legitimate process in a suspended state, replace its memory with your shellcode, and resume it.
 - **Verification:** Use **Process Hacker** or **Sysinternals Suite** to see if you can find the hidden shellcode inside the legitimate process.
- **Learning Goal:** Understand advanced memory forensics and how attackers stay invisible in a crowded system.

Weekly Deliverable: The Evasion & Stealth Report

1. **Detection Comparison:** A table showing the detection rate of a "Raw Payload" vs. an "Obfuscated Payload."
 2. **Bypass Documentation:** Screenshots of a successful AMSI bypass and the commands used.
 3. **LOLBAS Log:** List of binaries you tested and how they helped in the attack chain.
 4. **Security Recommendation:** How can a Blue Team detect "In-Memory" attacks? (e.g., "Enabling PowerShell Script Block Logging").
-



Red Team Outline

3 Months

Month 03 | Week 10: C2 Frameworks & Phishing Infrastructure

Objective: To learn how to deploy professional Command & Control (C2) infrastructure and execute targeted phishing campaigns to gain initial access to a corporate network.

Task 01: Deploying a C2 Framework (Sliver or Havoc)

- **Description:** Move away from simple reverse shells and deploy a professional C2 framework to manage your "implants."
- **Technical Requirements:**
 - **Setup:** Install and configure the **Sliver C2** (by BishopFox) or the **Havoc C2** framework on your attack server.
 - **Listener Configuration:** Set up an HTTP/HTTPS listener to receive incoming connections.
 - **Implant Generation:** Generate a basic "Beacon" or "Implant" and execute it on a target VM to establish a persistent connection.
- **Learning Goal:** Understand how professional attackers manage dozens of compromised systems from a single centralized dashboard.

Task 02: Weaponizing Documents & Payloads

- **Description:** Create files that look innocent but execute your C2 implant when opened.
- **Technical Requirements:**
 - **Macro-Based Payloads:** Create a Microsoft Word or Excel document with a malicious **Macro (VBA)** that downloads and executes your shellcode.
 - **LNK Files:** Create a malicious **Shortcut (.LNK)** file that uses PowerShell to execute a hidden command in the background.
 - **HTML Smuggling:** Create an HTML file that "assembles" a malicious payload on the user's browser to bypass email gateway filters.



Red Team Outline

3 Months

- **Learning Goal:** Learn how to deliver payloads through common business file formats without raising suspicion.

Task 03: Phishing Infrastructure with GoPhish

- **Description:** Set up a full-scale phishing platform to track who clicks your links and who submits their passwords.
- **Technical Requirements:**
 - **Installation:** Set up **GoPhish** (Open-Source Phishing Framework) on your server.
 - **Landing Pages:** Use the "Import" feature to clone a legitimate login page (e.g., Microsoft 365 or a company VPN portal).
 - **Email Templates:** Design a highly convincing email template using Social Engineering triggers (Urgency, Authority, or Fear).
- **Learning Goal:** Mastering the technical side of "Email Delivery" and "Tracking" which is the #1 entry point for data breaches globally.

Task 04: Domain Spoofing & SMTP Fundamentals

- **Description:** Learn how attackers bypass email security protocols like SPF, DKIM, and DMARC.
- **Technical Requirements:**
 - **Research:** Explain how **SPF**, **DKIM**, and **DMARC** records prevent unauthorized emails.
 - **Tooling:** Use **Swaks** (Swiss Army Knife for SMTP) to send a test email from your server and analyze why it lands in "Spam" vs. "Inbox."
 - **Typosquatting:** Find a domain similar to a target (e.g., [google.com](https://www.google.com) vs [go0gle.com](https://www.go0gle.com)) and explain how it can be used for "Homograph Attacks."
- **Learning Goal:** Understanding why "Brand Protection" and "Domain Integrity" are vital for modern organizations.

Weekly Deliverable: The Initial Access & C2 Report

1. **C2 Dashboard Screenshot:** Show your Sliver/Havoc dashboard with at least one active session.



Red Team Outline

3 Months

2. **Phishing Campaign Results:** A screenshot from GoPhish showing "Email Sent," "Email Opened," and "Link Clicked."
3. **Weaponization Analysis:** A step-by-step guide on how you created your malicious document.
4. **Security Recommendation:** How can employees be trained to spot your specific phishing email?

Month 03 | Week 11: Cloud Security & Attacking Modern Infrastructure

Objective: To identify and exploit misconfigurations in Cloud environments (AWS/Azure) and understand how to pivot from a web vulnerability to full cloud account takeover.

Task 01: Cloud Reconnaissance & S3 Bucket Leaks

- **Description:** Identify public-facing cloud assets that belong to a target organization but are often overlooked.
- **Technical Requirements:**
 - **Bucket Hunting:** Use tools like **S3Scanner**, **CloudEnum**, or **GrayHatWarfare** to find publicly accessible S3 buckets or Azure Blobs.
 - **Manual Analysis:** Use the AWS CLI (`aws s3 ls`) to check for "List" and "Read" permissions on discovered buckets.
 - **Data Harvesting:** Search for sensitive files (e.g., `.db`, `.ssh`, `.conf`, or source code) inside the buckets.
- **Learning Goal:** Understand that "Security by Obscurity" fails in the cloud; if it's public, it will be found.

Task 02: Exploiting SSRF to Access Metadata Services



Red Team Outline

3 Months

- **Description:** Use a Web Vulnerability (SSRF) to steal Cloud Instance credentials.
- **Technical Requirements:**
 - Identify an **SSRF (Server-Side Request Forgery)** vulnerability in a web application.
 - Query the **IMDS (Instance Metadata Service)** at `http://169.254.169.254/latest/meta-data/iam/security-credentials/` to retrieve temporary Access Keys, Secret Keys, and Session Tokens.
 - Configure your local AWS CLI with these stolen credentials to see what level of access you have gained.
- **Learning Goal:** Learn how a simple web bug can lead to a total compromise of the underlying cloud infrastructure.

Task 03: Attacking IAM (The "Active Directory" of the Cloud)

- **Description:** Abuse Identity and Access Management (IAM) permissions to escalate privileges within the cloud account.
- **Technical Requirements:**
 - **Enumeration:** Use **Pacu** (Exploitation Framework for AWS) or **CloudMapper** to map the permissions of the current user.
 - **Privilege Escalation:** Identify "Over-privileged" users (e.g., a user with `iam:CreateAccessKey`) and use them to gain Administrative control.
 - **Azure Specifics:** Research how **Managed Identities** in Azure can be abused if assigned to a compromised Virtual Machine.
- **Learning Goal:** Understand that "Least Privilege" is the most violated principle in Cloud Security.

Task 04: Container & Kubernetes Security (Docker Escapes)

- **Description:** If you land inside a Docker container, you must learn how to "break out" to the host machine.
- **Technical Requirements:**
 - **Container Enumeration:** Check for "**Privileged Mode**" (`--privileged`) or exposed Docker sockets (`/var/run/docker.sock`).



Red Team Outline

3 Months

- **Escape Techniques:** Use tools like **CDK** or manual exploits to escape the container and gain a shell on the host OS.
- **K8s API Hacking:** Identify exposed Kubernetes API servers or insecure Kubelets that allow unauthenticated command execution.
- **Learning Goal:** Mastering the security of modern "Microservices" and the risks of container misconfigurations.

Weekly Deliverable: The Cloud & Infrastructure Audit Report

1. **Cloud Asset Map:** A list of discovered buckets, IP ranges, and identified services.
2. **Metadata Exploitation Log:** Screenshots showing the transition from an SSRF bug to valid cloud credentials.
3. **IAM Privilege Path:** A diagram showing how you escalated from a "Dev" user to a "Cloud Admin."
4. **Security Hardening:** Provide 3 critical steps to secure a Cloud environment (e.g., "Enforcing IMDSv2").

Month 03 | Week 12: The Capstone Project (Full Attack Chain)

Objective: To execute a complete, end-to-end simulated cyber attack on a complex corporate network, starting from external reconnaissance and ending with full Domain Dominance and a professional Boardroom-ready report.

Task 01: The Scoping & Planning Phase

- **Description:** Before the attack, you must define the "Rules of Engagement" (RoE).
- **Technical Requirements:**
 - Create a **Pre-Engagement Document** that outlines the scope, target IP ranges, and the "Flags" (Objectives) to be captured.



Red Team Outline

3 Months

- Define the "**Success Criteria**" (e.g., "Gaining access to the HR Database").
- **Learning Goal:** Learn that professional hacking is about planning and legal boundaries, not just clicking buttons.

Task 02: Execution of the "Cyber Kill Chain"

- **Description:** Perform a continuous attack from the outside in.
- **Technical Requirements:**
 - **Step 1:** External Recon & Phishing (Week 1, 10).
 - **Step 2:** Initial Access via Web Exploit or Credential Theft (Week 3, 4).
 - **Step 3:** Bypassing EDR/AV on the target host (Week 9).
 - **Step 4:** Privilege Escalation & Persistence (Week 5).
 - **Step 5:** Pivoting and Lateral Movement to the AD Environment (Week 6, 7).
 - **Step 6:** Reaching Domain Admin status (Week 8).
- **Learning Goal:** Prove that you can connect all the "Individual Skills" into one powerful, unstoppable attack flow.

Task 03: Post-Engagement Cleanup

- **Description:** A professional Red Teamer never leaves a mess behind.
- **Technical Requirements:**
 - Identify every "Backdoor" or "Persistence Mechanism" you created during the week.
 - Document the process of **reverting changes** (e.g., deleting temporary users, removing cron jobs, or cleaning up logs).
 - Verify that the target system is back to its original state.
- **Learning Goal:** Ethics and Professionalism. Leaving shells behind is a massive security risk for the client.

Task 04: The "Boardroom" Final Report

- **Description:** Create the final document that proves your value to the organization.



Red Team Outline

3 Months

- **Technical Requirements:**
 - **Executive Summary:** A high-level view for CEOs (Non-technical).
 - **Technical Breakdown:** Step-by-step documentation with screenshots for IT Admins.
 - **Risk Rating:** Categorize findings using **CVSS** and **MITRE ATT&CK** mapping.
 - **Strategic Remediation:** Provide long-term solutions, not just "quick fixes."
- **Learning Goal:** Understand that in the industry, **The Report is the Product.**

Weekly Deliverable: The Cyberster Graduation Report

Requirement: Submit a comprehensive 30+ page PDF report detailing the entire 12-week journey's climax. This report will be the primary factor for the **Cyberster Certificate of Excellence.**



Blue Team Outline

3 Months

Month 1: SOC Operations & SIEM Foundations

Week 1 — Lab Setup & SOC Foundations (Days 1–7)

Objective: To establish a professional defensive laboratory environment and master the fundamentals of network traffic analysis and SIEM architecture.

- **Day 1: Networking Fundamentals for SOC**
 - **Tasks:** * Review TCP/IP, OSI model, subnetting, and common ports.
 - Understand packet flow: ARP, DNS, HTTP, HTTPS, ICMP.
 - Map traffic paths in VirtualBox (NAT vs Bridged vs Host-Only).
 - Capture live traffic with Wireshark and identify protocols.
 - **Deliverables:** Network diagram of lab topology, Wireshark capture with labeled protocols, and short notes on key protocols.
 - **Learning Goal:** Build a strong foundation in network communications to differentiate between normal and malicious traffic.
- **Day 2: VirtualBox Lab Setup**
 - **Tasks:** * Install VirtualBox & create 3 VMs: Kali Linux, Ubuntu Server (Wazuh), and Windows 10.
 - Configure Host-Only + NAT adapters for correct routing.
 - Verify connectivity between all VMs and take VM snapshots.
 - **Deliverables:** Detailed report with screenshots of all 3 VMs running, network connectivity verification output, and snapshot confirmation.
 - **Learning Goal:** Master the creation of an isolated, safe sandbox environment for security testing.
- **Day 3-4: Wazuh Manager Installation**
 - **Tasks:** * Install Wazuh Manager + Indexer + Dashboard on Ubuntu Server VM.
 - Verify services: wazuh-manager, wazuh-indexer, wazuh-dashboard.
 - Access Wazuh dashboard at <https://<IP>:443> and explore tabs (Overview, Agents, Security Events).



Blue Team Outline

3 Months

- **Deliverables:** Detailed report with screenshots of every step, service status output (systemctl status), and IP configuration screenshot.
- **Learning Goal:** Understand SIEM architecture and the server-side components of defensive monitoring.
- **Day 5-6: Wazuh Agent Deployment**
 - **Tasks:** * Deploy Wazuh agent on Kali Linux and Windows 10 VMs.
 - Configure agent to connect to Wazuh Manager via ossec.conf.
 - Verify agent appears as "Active" in the dashboard.
 - Generate test alerts by triggering failed SSH logins.
 - **Deliverables:** Detailed report with screenshots of every step, agent configuration file (ossec.conf), and initial alert screenshots.
 - **Learning Goal:** Learn the client-server relationship in endpoint monitoring and how telemetry is collected.
- **Day 7: Wazuh Dashboard Navigation & Alert Triage**
 - **Tasks:** * Explore: Security Alerts, Threat Hunting, and MITRE ATT&CK modules.
 - Filter alerts by rule level (0-15), agent, and time range.
 - Document the top 10 alert types seen in the environment.
 - **Deliverables:** Alert triage report, screenshots of major dashboard modules, and top 10 alerts summary.
 - **Learning Goal:** Transition from setup to analyst operations by learning how to navigate and filter security data.

Week 2 — File Integrity Monitoring & Rule Tuning (Days 8–14)

Objective: To implement advanced host-based monitoring and customize detection logic to reduce noise and identify specific threats.



Blue Team Outline

3 Months

- **Task 01: Wazuh FIM Monitoring**
 - **Description:** Configure Wazuh File Integrity Monitoring (FIM) to monitor critical system directories and generate real-time alerts on unauthorized file changes.
 - **Technical Requirements:** * Configure FIM in ossec.conf to monitor: `/etc`, `/var/log`, and `C:\Windows\System32`.
 - Set syscheck frequency and enable real-time monitoring.
 - Create, modify, and delete test files to observe alerts.
 - Identify "who, what, and when" from FIM events.
 - **Deliverables:** Detailed report of every step with screenshots of FIM alerts in Wazuh dashboard, ossec.conf FIM configuration, and FIM event analysis report.
 - **Learning Goal:** Understand how to detect unauthorized system changes and tampering—a key indicator of a compromise.
- **Task 02: Custom Wazuh Rules & Log Analysis**
 - **Description:** Study the Wazuh rule structure and create custom detection logic for specific organizational threats.
 - **Technical Requirements:** * Write 3 custom rules: detect new user creation, SSH brute force, and USB insertion.
 - Analyze `/var/log/auth.log` and Windows Event Logs (Event IDs 4625, 4672, 4720).
 - Tune false positives by adjusting rule thresholds.
 - **Deliverables:** Custom rules XML file, screenshots of triggered alerts, and a log analysis report with a timeline of suspicious activity.
 - **Learning Goal:** Learn to customize a SIEM to your specific environment and master the manual analysis of raw logs.
- **Task 03: Wazuh Active Response**
 - **Description:** Configure automated defensive actions to stop attacks in real-time.



Blue Team Outline

3 Months

- **Technical Requirements:** * Configure active response to auto-block IPs in the firewall after 5 failed SSH attempts.
 - Execute a brute-force attack from Kali to the Ubuntu server to trigger the block.
 - Document the active response workflow and verify the firewall rule application.
- **Deliverables:** Screenshot of blocked IP in iptables, active response configuration snippet, and alert showing the auto-block trigger.
- **Learning Goal:** Experience the transition from passive monitoring to automated active defense.

Weekly Deliverable: The SIEM Engineering Report

- **FIM Analysis:** Documentation of file changes detected on critical hosts.
- **Custom Rule Set:** The XML code and logic behind your custom detection rules.
- **Active Defense Log:** Proof of the automated blocking of a simulated attacker.
- **Security Recommendation:** How can an organization balance between "Real-time monitoring" and "System performance"?

Month 2: Network IDS & Advanced Monitoring

Week 3 — Suricata IDS Integration with Wazuh (Days 15–21)

Objective: To transition from host-based to network-based detection by integrating Suricata IDS for deep packet inspection.

- **Task 01: Suricata Installation & Custom Rule Writing**
 - **Description:** Deploy and configure Suricata as a Network Intrusion Detection System (NIDS).



Blue Team Outline

3 Months

- **Technical Requirements:** * Install Suricata on Kali and configure suricata.yaml for live traffic capture.
 - Write 3 custom rules: detect Nmap scans, HTTP exploit attempts, and ICMP floods.
 - Reload rules and verify alerts in fast.log and eve.json.
- **Deliverables:** Suricata configuration files, custom rules file, and screenshots of alerts triggered in the raw logs.
- **Learning Goal:** Understand signature-based network detection and how to write custom NIDS signatures.
- **Task 02: Suricata + SIEM Integration**
 - **Description:** Bridge the gap between network logs and the SIEM dashboard.
 - **Technical Requirements:** * Configure the Wazuh agent to monitor Suricata's eve.json output.
 - Perform port scanning and malicious URL requests (wget) to generate traffic.
 - Observe Suricata alerts flowing into the Wazuh dashboard.
 - **Deliverables:** Screenshot showing Suricata alerts in the Wazuh dashboard and a correlation analysis of combined host/network alerts.
 - **Learning Goal:** Learn the value of "Log Aggregation"—seeing network alerts and host alerts in a single pane of glass.

Week 04 — pfSense Firewall & Advanced Monitoring (Days 22–28)

Objective: To secure the perimeter using a professional-grade firewall and utilize threat intelligence to proactively hunt for indicators of compromise (IOCs).

- **Task 01: pfSense Firewall Setup & SIEM Integration**



Blue Team Outline

3 Months

- **Description:** Install and configure a network firewall to act as the primary gateway for your lab environment.
- **Technical Requirements:**
 - Install **pfSense** as a VM gateway in VirtualBox.
 - Route all lab VM traffic through pfSense and configure WAN/LAN interfaces.
 - Create firewall rules to restrict traffic and enable detailed logging.
 - Connect pfSense syslog output to the **Wazuh Manager** for centralized monitoring.
- **Learning Goal:** Understand perimeter defense and how to aggregate network-level security logs into a SIEM.
- **Task 02: Threat Intelligence Integration**
 - **Description:** Enhance your detection capabilities by integrating real-world threat feeds and intelligence modules.
 - **Technical Requirements:**
 - Configure the **Wazuh VirusTotal** integration to automatically check file hashes against global databases.
 - Enable and explore the **MITRE ATT&CK** module to map incoming alerts to known adversary techniques.
 - Integrate open-source threat feeds such as **URLhaus**, **Abuse.ch**, or **Emerging Threats**.
 - **Learning Goal:** Learn how "Intelligence-Driven Defense" helps SOC analysts stay ahead of evolving threats.
- **Task 03: Vulnerability Assessment with Wazuh**
 - **Description:** Proactively identify weaknesses in your systems before an attacker can exploit them.
 - **Technical Requirements:**
 - Enable the **Wazuh Vulnerability Detection** module on both Windows and Linux agents.



Blue Team Outline

3 Months

- Analyze the assessment results to identify critical **CVEs** on installed packages.
- Prioritize vulnerabilities based on their **CVSS score** (Low to Critical).
- Simulate a patching process and re-scan the system to verify remediation.
- **Learning Goal:** Master vulnerability management, which is a core responsibility of any defensive security team.
- **Task 04: SOC Dashboards & Executive Reporting**
 - **Description:** Convert raw security data into meaningful visualizations and professional reports for leadership.
 - **Technical Requirements:**
 - Build a custom **Wazuh Dashboard** featuring: alert counts by severity, agent health status, top triggered rules, and a geolocation map of attacks.
 - Generate a weekly **SOC Summary Report** documenting significant events.
 - Practice writing an **Executive Summary** that translates technical data into business risk.
 - **Learning Goal:** Understand that data visualization and clear communication are vital for making security decisions at the corporate level.

Weekly Deliverable: The Perimeter & Vulnerability Audit

- **pfSense Configuration:** Screenshot of the firewall dashboard showing active traffic and rules.



Blue Team Outline

3 Months

- **Vulnerability Report:** A 3-4 page document listing discovered CVEs, their impact, and prioritized remediation steps.
- **Custom SOC Dashboard:** A high-quality screenshot of your personalized monitoring interface.
- **Executive Summary:** A 1-page summary of the week's security posture for "Board-level" review.

Week 5 — Malware Analysis & Incident Response (Days 29–35)

Objective: To analyze malicious software and execute a structured incident response plan following the NIST framework.

- **Task 01: Static & Dynamic Malware Analysis**
 - **Description:** Safely analyze malware samples to extract Indicators of Compromise (IOCs).
 - **Technical Requirements:** * Perform static analysis: Use `strings`, `exiftool`, and `binwalk` to find hardcoded IPs and hashes.
 - Perform dynamic analysis: Upload samples to **ANY.RUN** or a local sandbox to observe process spawning and C2 connections.
 - Upload hashes to VirusTotal and document detection rates.
 - **Deliverables:** Static analysis report per sample, IOC list (hashes, IPs, domains), and process tree documentation from ANY.RUN.
 - **Learning Goal:** Master the methodology of dissecting malware to understand its intent and behavior.
- **Task 02: Incident Response Planning**
 - **Description:** Create a professional response plan for a simulated malware infection.
 - **Technical Requirements:** * Identify Indicators of Attack (IOAs) from Wazuh alerts during the malware execution.



Blue Team Outline

3 Months

- Write a full IR Plan using the NIST framework: Preparation, Detection, Containment, Eradication, and Recovery.
- Document the remediation steps (e.g., "Kill process X," "Delete registry key Y").
- **Deliverables:** Incident Response Plan document (5-7 pages) and a summary table of IOCs/IOAs.
- **Learning Goal:** Understand that technology is only half the battle; a structured process is required to handle a breach professionally.

Week 06: Advanced Malware Analysis & SOC Capstone (Days 36–45)

Objective: To master hybrid analysis techniques and demonstrate full-spectrum defensive capabilities through a simulated multi-stage breach.

- **Task 01: Advanced Malware Analysis (Task 04)**
 - **Description:** Move beyond basic executable analysis to investigate document-based malware and sophisticated delivery mechanisms.
 - **Technical Requirements:**
 - Prepare a Windows 7 "Detonation" VM and select a DOCX/RTF-based malware sample (e.g., *Guinea.rtf*).
 - Perform hybrid analysis: static metadata extraction plus dynamic sandbox execution.
 - Analyze LNK persistence, dropped DLLs, and macro-based exploits.
 - Cross-reference findings with professional reports from *Hybrid-Analysis.com*.



Blue Team Outline

3 Months

- **Learning Goal:** Understand how attackers use legitimate document formats to bypass traditional email filters.
- **Task 02: Threat Hunting in Wazuh**
 - **Description:** Proactively search for hidden threats within the SIEM that automated rules might have missed.
 - **Technical Requirements:**
 - Use the Wazuh Threat Hunting module to hunt for LOLBins (Living off the Land Binaries) and unusual parent-child process relationships.
 - Identify lateral movement indicators and unusual outbound network connections.
 - Create and save custom hunting queries and rules in Wazuh.
 - **Learning Goal:** Shift from a reactive mindset (waiting for alerts) to a proactive defensive posture.
- **Task 03: Major Breach Case Studies**
 - **Description:** Analyze real-world 2024-2025 breaches (e.g., Snowflake, AT&T) to understand modern TTPs.
 - **Technical Requirements:**
 - Identify attack vectors, IOCs, and MITRE techniques for 5 major recent breaches.
 - Determine which Wazuh rules or Suricata signatures could have detected these attacks.
 - **Learning Goal:** Bridge the gap between lab training and the actual threat landscape.
- **Task 04: SOC Phase Capstone Project**
 - **Description:** Execute a full "Red vs. Blue" simulation involving brute force, privilege escalation, and data exfiltration.
 - **Technical Requirements:**
 - Simulate a full attack chain and perform real-time triage, containment, and reporting.
 - Present findings in a 15-minute SOC debrief presentation.



Blue Team Outline

3 Months

- **Learning Goal:** Connecting all Phase 1 skills into a single, unstoppable defensive workflow.

Weekly Deliverable: The SOC Capstone Report

- **Full Incident Report:** Documentation of the capstone attack detection and response.
- **MITRE Mapping Table:** TTPs identified during the threat hunt and case studies.
- **SOC Debrief Slides:** A professional presentation (10-12 slides) of findings.

Week 07: DFIR Foundations — Disk Imaging & File Systems (Days 46–52)

Objective: To learn the legal and technical requirements for evidence preservation and file system architecture.

- **Task 01: Forensic Disk Imaging & Hash Verification**
 - **Description:** Acquire digital evidence in a forensically sound manner.
 - **Technical Requirements:**
 - Create a forensic image (E01 format) of a USB drive using **FTK Imager**.
 - Generate and verify MD5/SHA-256 hashes to ensure data integrity.
 - Mount the image as "Read-Only" using **Arsenal Image Mounter**.
 - Document the "Chain of Custody" for the evidence.
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Blue Team Outline

3 Months

- **Learning Goal:** Understand that the integrity of evidence is the foundation of any forensic investigation.
- **Task 02: File System Analysis & MFT Examination**
 - **Description:** Dive deep into how operating systems store and hide data.
 - **Technical Requirements:**
 - Use **MFT Explorer** to examine Master File Table entries in NTFS.
 - Create and detect files hidden in **Alternate Data Streams (ADS)**.
 - Document timestamp changes (MACB times) following file operations (Modify, Access, Create, Birth).
 - **Learning Goal:** Learn to identify "anti-forensic" techniques like timestamp manipulation and hidden streams.
- **Task 03: Autopsy Case Setup & Initial Triage**
 - **Description:** Initialize a professional forensic case and run automated ingest modules.
 - **Technical Requirements:**
 - Configure an **Autopsy** case and add an E01 image.
 - Run ingest modules: Keyword search, EXIF data, and File type identification.
 - Identify and "carve" deleted files from unallocated space.
 - **Learning Goal:** Master the primary open-source tool used for deep-dive forensic analysis.

Week 08: Windows Artifacts — Event Logs & Prefetch (Days 53–59)

Objective: To reconstruct user activity and program execution history through system-level artifacts.

- **Task 01: Windows Event Log Deep Dive**



Blue Team Outline

3 Months

- **Description:** Analyze authentication and system logs to identify malicious behavior.
- **Technical Requirements:**
 - Identify successful (4624) and failed (4625) logons.
 - Detect brute force patterns and privilege escalation events (4672).
 - Track account management activities (4720 - User Creation).
- **Learning Goal:** Translate raw Event IDs into a coherent narrative of what happened on a system.
- **Task 02: Program Execution Analysis (Prefetch)**
 - **Description:** Prove that specific programs were run, even if they have been deleted.
 - **Technical Requirements:**
 - Extract prefetch files and parse them using **PECmd**.
 - Identify programs executed from suspicious locations (Temp, Downloads).
 - Build an execution timeline for the past week.
 - **Learning Goal:** Use execution artifacts to prove "intent" and identify the use of malicious tools.

Month 3: Digital Forensics & Investigation

Week 09: Browser Forensics & LNK File Analysis (Days 60–66)



Blue Team Outline

3 Months

Objective: To track a user's internet activity and their interaction with files and external devices.

- **Task 01: Browser Forensics (Chrome, Firefox, Edge)**
 - **Description:** Extract browsing history, downloads, and session data.
 - **Technical Requirements:**
 - Analyze SQLite databases for history and cookies.
 - Extract download origins and timestamps.
 - Document installed extensions and their associated permissions.
 - **Learning Goal:** Reconstruct the suspect's research and external communications.
- **Task 02: LNK & Recent File Analysis**
 - **Description:** Identify which files were opened and which USB devices were connected.
 - **Technical Requirements:**
 - Parse LNK files using **LECcmd** to find volume serial numbers.
 - Identify network share access and server names in LNK metadata.
 - Create a chronological timeline of file access events.
 - **Learning Goal:** Connect a user to a specific file or an external "leak" device.

Week 10: Mantooth Labs — Advanced Autopsy Investigation (Days 67–73)

Objective: To apply forensic skills to a complex, real-world case scenario.

- **Task 01: Mantooth Lab 1 — Initial Triage**
 - **Description:** Perform the first stage of the Mantooth forensic challenge.
 - **Technical Requirements:**



Blue Team Outline

3 Months

- Extract computer name, OS version, and active users from the image.
 - Analyze user activity: recent documents, browser history, and downloads.
 - Answer challenge questions with specific evidence citations.
- **Learning Goal:** Practice systematic triage under the pressure of a specific case objective.
- **Task 02: Mantooth Lab 2 — EZ Tools Deep Dive**
 - **Description:** Use advanced command-line tools to correlate disparate artifacts.
 - **Technical Requirements:**
 - Parse **Jump Lists (JLECmd)** and **ShellBags (SBECmd)** to identify folder access patterns.
 - Build a unified timeline from all sources using **EZ Timeline**.
 - Identify "Cross-Artifact" correlations (e.g., a Jump List entry matching a Prefetch execution).
 - **Learning Goal:** Learn how to use multiple artifacts to confirm a single event, making your evidence "court-room ready."

Week 11: Registry Forensics (Days 74–80)

Objective: To master the "Heart of the OS"—the Windows Registry—to uncover system configuration and hidden user activity.

- **Task 01: System & SAM Hive Analysis**
 - **Description:** Extract hardware history and user account metadata.



Blue Team Outline

3 Months

- **Technical Requirements:**
 - Analyze **SYSTEM hive** for USBSTOR (USB history) and Timezone settings.
 - Analyze **SAM hive** for account creation dates, RID values, and last login times.
- **Learning Goal:** Uncover "persistence" at the system level and map out the user environment.
- **Task 02: Software & User Hive (NTUSER.DAT) Analysis**
 - **Description:** Analyze installed software and personal user activity.
 - **Technical Requirements:**
 - Document **AutoStart** locations (Run/RunOnce keys) to find malware persistence.
 - Decode **UserAssist** (ROT-13 encoded data) to find execution counts.
 - Parse **ShellBags** to reconstruct the folders the user viewed in Explorer.
 - **Learning Goal:** See exactly what the user saw and what they intended to hide.

Weekly Deliverable: The Comprehensive Registry Report

- **Registry Inventory:** A 25-30 page report covering USB history, persistence, and user activity.
- **Decoded Artifacts:** Decoded UserAssist and ShellBag timelines.
- **Evidence Presentation:** A 10-15 slide summary for technical leads.

Week 12 — The M57.biz Capstone (Final Investigation)



Blue Team Outline

3 Months

Objective: To execute a full-scale forensic investigation, correlating email, registry, and disk artifacts to solve a complex corporate espionage case.

- **Task 01: System Analysis & User Profiling**

- **Description:** Reconstruct the environment and the suspect's role.
- **Technical Requirements:** * Load the case images into Autopsy and run ingest modules.
 - Determine the suspect's job role through email and document analysis.
 - Extract system info: Computer name, OS version, and user accounts.
- **Deliverables:** System analysis report and user profile document.
- **Learning Goal:** Learn to build a "contextual profile" of a suspect before diving into deep technical analysis.

- **Task 02: Evidence Correlation & Super Timeline**

- **Description:** Build a master timeline that connects all dots.
- **Technical Requirements:** * Collect timestamps from: File system, Email (PST/OST), Browser history, Registry, and Prefetch.
 - Correlate events: "Email received" -> "File downloaded" -> "USB inserted" -> "File copied."
 - Map findings to the MITRE ATT&CK framework (Exfiltration, Collection).
- **Deliverables:** Complete Super Timeline (CSV/Excel) and a visual timeline with narrative annotations.
- **Learning Goal:** Prove the "Chain of Events" required for legal or corporate action.

- **Task 03: Final Capstone Presentation**

- **Description:** Present your findings to a "Board of Directors."
- **Technical Requirements:** * Create a 15-20 slide professional presentation.



Blue Team Outline

3 Months

- Organize an evidence package with numbered screenshots and hashes.
- Complete a Chain of Custody document.
- **Deliverables:** Final Forensic Report (30+ pages), Capstone Presentation, and Evidence Package.
- **Learning Goal:** Understand that in forensics, your ability to explain your findings is as important as the findings themselves.

Weekly Deliverable: The Cyberster Blue Team Graduation Report

- **Requirement:** Submit a comprehensive PDF report detailing the SOC and DFIR journey's climax. This report will be the primary factor for the **Cyberster Certificate of Excellence**.
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Purple Team Outline

3 Months

Month 1: Network Security & Protocol Analysis

Week 01: Network Security Lab Foundations

Objective: To initialize a professional hybrid laboratory environment and master the mechanics of low-level network communication.

- **Task 01: Virtualization & Network Topology Setup**
 - **Description:** Build the foundation of the Purple Team lab by configuring isolated attack and defense segments.
 - **Technical Requirements:**
 - Initialize Virtualization Software (VirtualBox/VMware) with dedicated Attacker (Kali) and Victim (Windows/Linux) machines.
 - Configure a strictly "Internal Network" to prevent accidental leakage while allowing inter-VM communication.
 - Verify connectivity using static IP assignments to ensure a stable testing environment.
 - **Learning Goal:** Master the creation of a safe, controlled sandbox necessary for executing high-risk security simulations.
- **Task 02: Packet Analysis & Protocol Deep-Dive**
 - **Description:** Use industry-standard tools to visualize how data moves across the wire.
 - **Technical Requirements:**
 - Perform live Packet Analysis via **Wireshark** to observe background network chatter.
 - Analyze and document the **TCP Three-Way Handshake** (SYN, SYN-ACK, ACK) in a live capture.
 - Identify common protocol headers (Ethernet, IP, TCP/UDP) and their forensic significance.
 - **Learning Goal:** Understand the fundamental "language" of the network to later identify when those protocols are being abused.



Purple Team Outline

3 Months

Week 02: Network Discovery & Reconnaissance

Objective: To contrast aggressive scanning techniques with stealthy discovery and implement the first layer of defensive detection.

- **Task 01: Active & Passive Discovery**
 - **Description:** Map the network using both "loud" and "silent" methodologies.
 - **Technical Requirements:**
 - Perform Active Discovery using ARP/ICMP sweeps to identify live hosts.
 - Execute a **TCP SYN Scan** (Stealth Scan) and compare its fingerprint to a Full Connect Scan.
 - Conduct Service Identification to determine versions of running applications (Fingerprinting).
 - **Learning Goal:** Understand the trade-off between speed of discovery and the likelihood of being detected by a SOC.
- **Task 02: Defensive Summary & IDS Implementation**
 - **Description:** Analyze the "Attacker" footprints from a defensive perspective.
 - **Technical Requirements:**
 - Use **Wireshark** to analyze the specific packet patterns generated by Nmap's stealth scans.
 - Implement a basic IDS/Defensive Summary to alert on high-frequency connection attempts.
 - **Learning Goal:** Learn how signature-based systems catch scanners by identifying non-standard packet flags.

Week 03: Network Protocol Abuse



Purple Team Outline

3 Months

Objective: To simulate and detect availability-based attacks and traffic interception (Man-in-the-Middle).

- **Task 01: Denial of Service (DoS) Simulations**
 - **Description:** Understand how to disrupt service availability and how to monitor for such disruptions.
 - **Technical Requirements:**
 - Differentiate between DoS and DDoS architectures in a technical report.
 - Simulate a DoS attack against a vulnerable network service (e.g., SYN Flood or Resource Exhaustion).
 - **Learning Goal:** Identify the point of failure in network protocols when stressed beyond their limits.
- **Task 02: MITM & ARP Spoofing Analysis**
 - **Description:** Intercept traffic between two legitimate nodes on the local network.
 - **Technical Requirements:**
 - Execute an **ARP Spoofing** attack to position the attacker machine as the "Man-in-the-Middle."
 - Use Wireshark to capture and analyze redirected packets, identifying unencrypted credentials or data.
 - **Learning Goal:** Understand the inherent trust vulnerabilities in ARP and the importance of encryption (TLS/SSH).

Month 2: Lateral Movement & Defensive Engineering

Week 04: Lateral Movement & Internal Network Attacks

Objective: To simulate the post-exploitation phase where an attacker moves through the internal environment.



Purple Team Outline

3 Months

- **Task 01: SMB Protocol Abuse & Enumeration**
 - **Description:** Exploit weak internal configurations to gain further access.
 - **Technical Requirements:**
 - Perform Internal Enumeration of Shared Resources (Admin shares, IPC\$, etc.) after compromising the SMB protocol.
 - Analyze SMB traffic in Wireshark to identify cleartext authentication or sensitive file transfers.
 - **Learning Goal:** Understand why SMB is a primary target for internal movement and how it can be secured.
- **Task 02: Practical Lateral Movement**
 - **Description:** Expand the "attack surface" by pivoting from one compromised service to another.
 - **Technical Requirements:**
 - Practice moving laterally by utilizing credentials found on one service to access another on the same network.
 - Identify the artifacts left behind during a "Pass-the-Hash" or service compromise.
 - **Learning Goal:** Master the "pivot" technique used by advanced persistent threats (APTs) to reach the "crown jewels" of a network.

Week 05: Network Detection & Defense Engineering

Objective: To design, implement, and test custom detection signatures for modern network threats.

- **Task 01: IDS vs. IPS Logic**
 - **Description:** Determine where to place defensive appliances and how to configure their response actions.
 - **Technical Requirements:**



Purple Team Outline

3 Months

- Define the placement strategy for an IDS (Out-of-band) vs. an IPS (In-line).
- Deploy **Suricata** or **Zeek** in the lab environment to monitor cross-segment traffic.
- **Learning Goal:** Understand the operational impact of "Block" (IPS) vs. "Alert" (IDS) modes.
- **Task 02: Custom Detection Rule Creation**
 - **Description:** Transition from using default signatures to writing custom defense logic.
 - **Technical Requirements:**
 - Create practical detection rules for previously practiced attacks (e.g., SYN Floods or ARP Spoofing).
 - Verify the rules by re-running the attacks and confirming alerts in the management console.
 - **Learning Goal:** Learn the syntax of defense—converting an attacker's TTP (Tactics, Techniques, and Procedures) into a working signature.

Month 2: Scripting, Automation & Web Security

Week 06: Scripting & Automation in Network Security

Objective: To master the use of higher-level languages for automating repetitive security tasks and building custom Purple Team tools.

- **Task 01: Automation Fundamentals & Environmental Setup**
 - **Description:** Understand the role of automation in scaling both offensive operations and defensive monitoring.
 - **Technical Requirements:**



Purple Team Outline

3 Months

- Define the importance of scripting in modern SOC environments and Red Team engagements.
- Set up a Python and Bash development environment within the lab.
- Analyze sample automation tools and identify the libraries used (e.g., Scapy, Requests, or Subprocess).
- **Learning Goal:** Recognize how automation reduces human error and increases the speed of response/attack.
- **Task 02: Practical Security Scripting (Python & Bash)**
 - **Description:** Create functional scripts to automate network reconnaissance and log parsing.
 - **Technical Requirements:**
 - **Bash:** Create a script to automate the discovery of live hosts and open ports using local system tools.
 - **Python:** Develop a script to parse a web server log file and extract all unique IP addresses or identify potential SQLi attempts.
 - **Integration:** Use Python's Scapy library to create a custom packet crafter for testing IDS signatures.
 - **Learning Goal:** Gain the ability to "build your own tools" instead of relying solely on pre-built software.

Week 07: Web App Foundations & SQL Injection Mastery

Objective: To bridge the gap between network and application layers by mastering SQL Injection (SQLi).

- **Task 01: Burp Suite & Web Architecture**
 - **Description:** Set up a vulnerable web environment and master the industry-standard proxy tool.



Purple Team Outline

3 Months

- **Technical Requirements:**
 - Configure a Vulnerable Web Application (e.g., DVWA or Juice Shop).
 - Utilize **Burp Suite** (Interceptor, Repeater, Intruder) to manipulate HTTP requests.
- **Learning Goal:** Master the "Proxy-based" workflow essential for all web application testing.
- **Task 02: SQL Injection Practical Approaches**
 - **Description:** Exploit database vulnerabilities using various manual and automated methods.
 - **Technical Requirements:**
 - Manually exploit SQLi types: **Error-based, Union-based, Boolean-based, and Time-based.**
 - Use automation tools like **sqlmap** to increase the efficiency of data extraction.
 - **Learning Goal:** Understand how improper input validation leads to total database compromise.

Week 08: Cross-Site Scripting (XSS) & Client-Side Attacks

Objective: To understand the client-side execution model and how to exploit or defend against attacks targeting the user's browser.

- **Task 01: XSS Analysis & Exploitation**
 - **Description:** Identify and exploit vulnerabilities that allow execution of malicious scripts in a victim's browser.
 - **Technical Requirements:**
 - Identify the difference between **Reflected, Stored, and DOM-based XSS.**



Purple Team Outline

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- Perform a practical XSS exploitation to steal session cookies or redirect users to a malicious site.
- Analyze how modern browsers use Content Security Policy (CSP) to mitigate these risks.
- **Learning Goal:** Understand the risk of unvalidated user input appearing in the browser's DOM.
- **Task 02: CSRF & Client-Side Defenses**
 - **Description:** Exploit Cross-Site Request Forgery (CSRF) to perform unauthorized actions on behalf of a user.
 - **Technical Requirements:**
 - Clearly explain the difference between XSS (script execution) and CSRF (unauthorized action).
 - Execute a CSRF attack to change a victim's profile information or password.
 - Implement and test defensive measures such as Anti-CSRF tokens and SameSite cookie attributes.
 - **Learning Goal:** Master the logic of "State-Changing" attacks and the importance of secure session management.

Month 3: Web App, API, & Cloud Security

Week 09: API Security Fundamentals & Enumeration

Objective: To learn how to secure the backbone of modern applications—Application Programming Interfaces (APIs).

- **Task 01: API Architecture & Endpoint Discovery**



Purple Team Outline

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- **Description:** Analyze how APIs function and map out the available attack surface.
- **Technical Requirements:**
 - Differentiate between **REST** and **SOAP** architectures at a high level.
 - Analyze JSON-based API requests and responses using **Postman** or Burp Suite.
 - Perform API endpoint enumeration to find "hidden" or undocumented routes.
- **Learning Goal:** Understand how APIs transfer data and why they require distinct security considerations compared to traditional web pages.
- **Task 02: Exploiting IDOR Vulnerabilities**
 - **Description:** Test for Insecure Direct Object Reference (IDOR) flaws in API endpoints.
 - **Technical Requirements:**
 - Identify endpoints that use sequential or predictable IDs (e.g., `/api/user/101`).
 - Manipulate request parameters to access data belonging to other users.
 - Propose defensive fixes using UUIDs and proper server-side authorization checks.
 - **Learning Goal:** Recognize that authentication (who you are) does not equal authorization (what you can see).

Week 10: API Authentication Attacks & Advanced Testing

Objective: To exploit flaws in modern authentication protocols like JWT and OAuth.

- **Task 01: JWT & Token Manipulation**



Purple Team Outline

3 Months

- **Description:** Analyze and attack JSON Web Tokens (JWT) used for session management.
- **Technical Requirements:**
 - Deconstruct a JWT into its Header, Payload, and Signature.
 - Identify common vulnerabilities: "None" algorithm attack, weak secret keys, or lack of signature verification.
 - Perform a token manipulation attack to escalate privileges to "Admin" status.
- **Learning Goal:** Understand how stateless authentication can be compromised if tokens are not properly validated.
- **Task 02: OAuth & API Fuzzing**
 - **Description:** Test for misconfigurations in OAuth flows and perform automated fuzzing for hidden flaws.
 - **Technical Requirements:**
 - Test for OAuth redirect URI misconfigurations.
 - Conduct API fuzzing using custom wordlists to identify **Broken Authentication** or unhandled error states.
 - Analyze the impact of leaky API responses.
 - **Learning Goal:** Learn to identify logic flaws in complex authentication handshakes.

Week 11: Business Logic Flaws & Advanced Web Attacks

Objective: To think like an attacker to find flaws in application logic and server-side resource handling.

- **Task 01: Server-Side Request Forgery (SSRF) & XXE**
 - **Description:** Force the server to perform requests on your behalf or exploit XML parsers.



Purple Team Outline

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- **Technical Requirements:**
 - Identify and test for **SSRF** to scan internal ports from the web server.
 - Identify and test for **XML External Entity (XXE)** to read local system files (e.g., /etc/passwd).
 - Analyze the impact of these attacks on internal cloud metadata services.
- **Learning Goal:** Master attacks that turn the server against its own internal infrastructure.
- **Task 02: Business Logic & Insecure Deserialization**
 - **Description:** Identify flaws in the "rules" of the application and the way objects are handled in memory.
 - **Technical Requirements:**
 - Identify Business Logic Flaws (e.g., adding a negative quantity to a shopping cart or bypassing payment steps).
 - Analyze the risks of Insecure Deserialization in high-level languages.
 - Document real-world impact scenarios for these "logic-based" findings.
 - **Learning Goal:** Learn that not all vulnerabilities are "bugs"—some are flaws in how the application was designed to function.

Week 12: Cloud Security & AI/ML Security Basics

Objective: To modernize the Purple Team skillset by addressing Cloud misconfigurations and the emerging AI threat landscape.

- **Task 01: Cloud Infrastructure Assessment**
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Purple Team Outline

3 Months

- **Description:** Identify the most common "low-hanging fruit" in AWS/Azure/GCP environments.
- **Technical Requirements:**
 - Identify common cloud misconfigurations (e.g., public S3 buckets).
 - Analyze **IAM (Identity and Access Management)** policies for over-privileged users.
 - Perform a simple cloud security assessment against a mock cloud environment.
- **Learning Goal:** Master the "Shared Responsibility Model"—knowing what the provider secures vs. what the customer must secure.
- **Task 02: AI/ML Security Risks**
 - **Description:** Understand how Artificial Intelligence systems can be attacked or subverted.
 - **Technical Requirements:**
 - Execute a **Prompt Injection** attack against a sample LLM to bypass safety filters.
 - Research and explain Model Abuse and Data Leakage risks in an enterprise setting.
 - Document the defensive strategies for AI systems (e.g., adversarial robustness).
 - **Learning Goal:** Prepare for the next generation of security threats where the "victim" is an algorithm.

Weekly Deliverable: The Purple Team Graduation Report

Requirement: Submit a comprehensive PDF detailing a **Full Attack/Defense Chain**. You must show the "Attack" (screenshots/commands), the "Detection" (IDS logs/alerts), and the "Remediation" (the fix). This report is the primary factor for the **Cyberster Purple Team Certificate of Excellence**.



Purple Team Outline

3 Months

